

Cross-Architectural Activation Alignment in Frontier Language Models:

A Representational-Geometry Pilot, with an Operator-Theoretic Program It Motivates

Adam A. Bensaid*

Claude (Anthropic)[†]

May 6, 2026

Abstract

We propose an operator-theoretic framing of the mechanistic interpretability literature’s empirical regularity that sparse autoencoders recover monosemantic feature dictionaries from trained transformers: namely, that these features are concrete realizations of approximately invariant subspaces of the nonlinear forward operator, in the sense the classical theory of invariant subspaces studies in the linear setting. From the framing we derive a research program, *Cross-Operator Subspace Isomorphism* (COSI), whose strong claim is that compositional reasoning subspaces are isomorphic across sufficiently different architectures. We do *not* test the strong claim in this paper. We report a representational-alignment pilot: two pre-registered runs in which Procrustes alignment of PCA-projected last-token activation matrices, between architecturally different frontier transformers on matched compositional-evaluation prompts, produces residuals far below a row-permutation null (empirical $p < 0.001$ in every pre-registered cell). The pilot demonstrates that the infrastructure detects a real cross-model representational signal, and we report it in full. The pilot does not demonstrate that the aligned subspaces are operator-invariant, that the alignment isolates compositional structure from lexical or template structure, that it is robust to held-out prompts, that it exceeds simple lexical baselines, or that it survives the training-distribution confound common to all current frontier models. We name each gap explicitly, specify the controls and interventions that would close it (invariance-leakage measurement, lexical and template baselines, train/test Procrustes, CKA cross-validation, training-distribution-disjoint controls, causal cross-model intervention), and run first-pass versions of four of them on the pilot data. Linear CKA confirms strong global representational similarity (0.84–0.96). Train/test Procrustes splits show no overfitting. Invariance-leakage measurement shows the PCA subspaces are partially invariant under the forward operator (more invariant than random subspaces; absolute leakage 0.51–0.89). **The lexical-baseline study materially weakens the operator-theoretic interpretation: the LLM–LLM Procrustes residual (0.77) is indistinguishable from one of the LLMs aligned against a TF-IDF representation of the same prompts (0.76), and is *worse* than a TF-IDF vs. template-only alignment (0.61).** The pilot’s representational signal is

*Atlas Consulting & Technology Services, Toronto, ON. Correspondence: adam.a.bensaid@gmail.com.

[†]Sonnet 4.5 / Opus 4.6 instances, accessed via Claude Code over the period 2026-04 to 2026-05. Authorial contribution: experimental design, code, draft prose. Final responsibility for all claims rests with the corresponding human author.

real but does not exceed what shared lexical/template structure alone would produce. We present the operator-theoretic program as a research agenda the pilot motivates rather than a finding the pilot confirms. The contribution we offer is the bridge between mechanistic interpretability and operator theory, the operationalization of one consequence of the Platonic Representation Hypothesis, and an honest accounting of what the present evidence does and does not support.

1 Introduction

Over the past two years, the mechanistic interpretability program has converged on an unexpected empirical regularity. Sparse autoencoders (SAEs) trained on the residual stream of large transformers recover dictionaries of features that, individually, behave as discrete computational primitives [1, 2, 3, 4]. The features are interpretable across many examples; they activate on recognizable concepts; their ablation produces predictable behavioral changes. This is the case across model families, model sizes, training regimes, and tokenizers. The pattern is sufficiently robust that the *universality hypothesis*—the conjecture that different networks trained on similar tasks recover similar internal computations—has been advanced as a working principle of the field [5, 6, 7].

A separate line of work has documented the same regularity from the representational-geometry angle. Bansal et al. [8] showed that the lower layers of one model can be “stitched” to the upper layers of another with only a thin trainable interface, and that the stitched model performs nearly as well as either original; this implies a deep correspondence between the intermediate representations of independently trained networks. Kornblith et al. [9] established centered kernel alignment (CKA) as a similarity measure robust to the symmetries that should not matter for the comparison and showed that representations across models are far more similar than the sheer dimensionality of the representation spaces would predict at chance. Most recently, Huh et al. [10] formulated the *Platonic Representation Hypothesis*: that as models scale and capability increases, their representations converge toward a shared statistical model of the world.

What has been missing, in our reading, is the mathematical object that this empirical convergence corresponds to.

The Operator-Theoretic Bridge

We propose that the SAE feature dictionary is a concrete realization of the *approximately invariant subspace structure* of the nonlinear forward operator that constitutes a transformer’s inference pass. The classical theory of invariant subspaces of bounded linear operators on Hilbert spaces [11, 12, 13, 14] asks whether every such operator admits a non-trivial closed invariant subspace—that is, a subspace V such that $T(V) \subseteq V$. The question is open in the Hilbert space setting; Lomonosov [11] established the affirmative for operators commuting with a non-zero compact operator, and Enflo [12] showed counter-examples exist for general Banach spaces. The structural intuition is the older one: when an operator admits an invariant subspace, its action decomposes into restrictions to simpler sub-operators, and the geometry of the operator becomes legible through the geometry of its invariant subspaces.

Transformer forward passes are not bounded linear operators. They compose linear projections, attention computations, and pointwise nonlinearities. The classical theorems do not apply directly. We use the language of invariant subspaces in an *approximate* sense: a subspace V is approximately invariant under a (possibly nonlinear) map f on a representative sample if the projection of $f(V)$ back onto V leaves a residual that is small compared to the operator’s overall action. This is the operational content of the empirical finding that SAE features are stable across forward passes: they identify directions in activation space that the forward operator preserves approximately.

The mechanistic interpretability literature has been doing operator theory for two years without naming it as such. Our first contribution is to make the bridge explicit.

The Falsifiable Consequence

If the bridge holds—if SAE feature dictionaries are bases for approximately invariant subspaces of the forward operators of trained transformers—then a sharp empirical consequence follows. Two models M_A and M_B with different architectures, run on the same compositional probe set, should admit an orthogonal alignment between their respective approximately invariant subspaces, with Procrustes residual significantly below the chance level set by a permutation null that breaks the semantic correspondence between rows while preserving each model’s internal geometry. Performance comparability between architectures (the weak universality claim) is consistent with many internal mechanisms; representational isomorphism is not. The isomorphism is testable. The test can return a null. We pre-register both the protocol and the significance criterion before observing data.

This is the *Cross-Operator Subspace Isomorphism* (COSI) program, of which the present paper is the calibration step.

What This Paper Reports

We report a representational-alignment pilot in two pre-registered runs. We are explicit that the pilot does not test the strong operator-theoretic claim; it tests a weaker but still interesting sub-claim, with controls and interventions that would close the gap specified as an explicit research program (§7).

Phase 0: Phi-4-Reasoning-Plus, a dense pure-attention reasoning model in the Phi-3 architecture family [15], against Qwen3-4B, a dense pure-attention generalist model in the Qwen3 family [16], on $N = 96$ compositional-evaluation prompts drawn from a single political-economy domain. The observed Procrustes residuals are 0.61, 0.81, 0.81 at the three pre-registered layer fractions; the empirical permutation p -value is < 0.001 at every layer (the observed residual is below the 1st percentile of the $S = 1000$ permutation null distribution).

Phase 1: Phi-4-Reasoning-Plus against Qwen3-Next-80B-A3B-Thinking [16], a hybrid attention/state-space-model architecture, on $N = 600$ prompts split equally across mathematical, logical, and political-economy compositional-evaluation domains. The observed Procrustes residuals are 0.77, 0.91, 0.96 at the three layer fractions, with empirical $p < 0.001$ at every layer and at every per-domain stratification cell (three domains $\times$ three layers, $n = 200$ per cell, $S = 500$ permutation samples).

What These Results Do and Do Not Show

These results show that, on matched compositional-evaluation prompts, the PCA-projected last-token activation matrices of two architecturally different frontier transformers admit a substantially better orthogonal alignment than they do under random row permutation. This is a real cross-model representational signal. We report it in full.

These results do *not* yet show:

1. That the aligned PCA subspaces are approximately invariant under the forward operator in the sense the operator-theoretic framing requires. The paper defines approximate invariance as a leakage statistic but does not measure it; PCA top- k components capture variance, not necessarily invariance.
2. That the alignment isolates compositional reasoning structure from lexical, template, syntactic, or tokenizer-induced structure. The row-permutation null is too weak to discriminate these. Two competent language models given the same English prompts will produce more aligned matched-row activations than scrambled-row activations *regardless* of any deeper substrate similarity.
3. That the alignment is robust to held-out probes. We fit Procrustes on the full probe set and evaluate on the same set; we do not perform a train/test split.
4. That the alignment exceeds the alignment achievable by simple lexical features (TF-IDF, sentence-transformer embeddings).
5. That the alignment survives the shared training-distribution confound. Both models in each pair are trained on overlapping post-2024 web corpora.
6. That the aligned subspace is causally responsible for any compositional behavior. Procrustes alignment is correspondence, not causation.

We are explicit about each of these gaps because the dramatic appearance of the headline numbers (residuals near 0.8 against null means near 1.0, with empirical $p < 0.001$ at every cell) could be read as much stronger evidence for the strong operator-theoretic claim than the experiment supports. Section 7 specifies the additional studies that would, jointly, support the strong claim. None of those studies has been run in the present paper.

Contributions

1. **An operator-theoretic framing as a research program.** We propose, and articulate at the level of the synthesis paper outlined elsewhere in this research program, that the mechanistic interpretability literature’s empirical regularities are most parsimoniously read as evidence for approximately invariant subspaces of the forward operator. The framing is a program, not a finding.
2. **Pre-registered methodology and infrastructure.** We specify a pre-registered protocol for cross-model representational alignment, with declared layer fractions, pooling,

dimensionality threshold, significance criterion, and null distributions. The infrastructure is released for reuse and extension.

3. **A representational-alignment pilot result.** Phases 0 and 1 demonstrate that matched compositional-evaluation prompts produce more orthogonally alignable PCA-projected activation geometries across architecturally different frontier transformers than under random row permutation, with empirical $p < 0.001$ in every pre-registered cell.
4. **An explicit accounting of what the pilot does not establish, and the studies that would close each gap.** Section 7 catalogues five required additional studies (invariance-leakage measurement, lexical and template baselines, train/test Procrustes with cross-domain transfer, training-distribution-disjoint controls, causal cross-model intervention) and indicates which of them addresses which gap.

Where This Sits in the Field

We take the operator-theoretic framing seriously enough to claim it unifies several open problems in interpretability that have been discussed largely in isolation: the universality hypothesis [5, 6], the polysemanticity / superposition problem [17], the cross-model transfer-of-mechanism question implicit in the Platonic Representation Hypothesis [10], and the substrate-independence intuition that has organized the broader cognitive science of representation since Marr [18]. We map these explicitly in Section 3. The bridge to computational neuroscience—to the neural-population doctrine [19] and to the geometry of biological invariants [20, 21, 22, 23]—and to the cybernetic and free-energy frameworks [24, 25, 26] that anticipate the same structure from independent grounds, we touch in the discussion (Section 9) but do not develop in this paper. The full integration is the subject of a forthcoming synthesis paper from the same research program.

Outline

Section 2 surveys the four literatures the bridge draws on: operator theory, mechanistic interpretability, cross-model representational alignment, and the substrate-independence tradition. Section 3 maps the open problems COSI sits among. Section 4 formalizes the COSI hypothesis and specifies the protocol. Section 5 reports the Phase 0 calibration. Section 6 catalogues the threats to validity that Phase 0 does not address. Section 9 specifies Phases 1–3 and articulates the implications, conditional on confirmation, across the literatures the framework bridges.

2 Background and Related Work

2.1 Invariant Subspaces of Bounded Operators

Let T be a bounded linear operator on a separable Hilbert space H . A closed subspace $V \subseteq H$ is *invariant* under T if $T(V) \subseteq V$. The *invariant subspace problem* asks whether every such T on an infinite-dimensional separable H admits a non-trivial closed invariant subspace—one other than $\{0\}$ or H itself. The question is open in the Hilbert space case; in finite dimensions every

operator trivially admits invariant subspaces (its eigenspaces, in particular). Two structural facts about the literature matter here.

First, when an operator admits an invariant subspace, its action decomposes: writing T in block form with respect to the splitting $H = V \oplus V^\perp$ shows that the upper-right block vanishes and the operator’s geometry becomes legible through the geometry of its restrictions to invariant subspaces. The existence of invariant subspaces is the existence of a useful coordinate system for the operator.

Second, the affirmative results form a hierarchy. Aronszajn and Smith [14] established that completely continuous (compact) operators always admit non-trivial invariant subspaces. Lomonosov [11] extended this dramatically: any operator commuting with a non-zero compact operator admits a non-trivial invariant subspace. Read [27] constructed counter-examples on ℓ^1 , and Argyros and Haydon [28] produced a Banach space on which every bounded operator has the form scalar-plus-compact (so every operator has invariant subspaces, but the construction requires extreme hereditary indecomposability). The Hilbert space case remains open. Beauzamy’s monograph [13] gives the standard treatment.

Transformer forward passes are not bounded linear operators. They are nonlinear maps composed of linear projections, multi-head attention (itself a nonlinear function of the input through softmax), gated feed-forward layers, and residual connections. The classical theorems do not apply. We use the language of invariant subspaces in an *approximate, empirical* sense: a k -dimensional subspace $V \subseteq \mathbb{R}^d$ is approximately ϵ -invariant under a map $f : \mathbb{R}^d \rightarrow \mathbb{R}^d$ on a sample X if $\|(I - P_V)f(P_V X)\|_F / \|f(P_V X)\|_F \leq \epsilon$, where P_V is the orthogonal projection onto V . This is a sample-level operationalization of the structural intuition the classical theory formalizes; we make no claim about the limiting infinite-dimensional case.

2.2 Mechanistic Interpretability: Circuits and Sparse Autoencoders

The mechanistic interpretability program seeks to identify the computational primitives implemented by trained neural networks, typically by reverse-engineering specific circuits or by decomposing the residual stream into interpretable components.

The circuits-level work has identified increasingly specific mechanisms. Olah et al. [5] articulated the program through case studies in vision models. Elhage et al. [29] provided a mathematical framework for transformer circuits. Olsson et al. [30] identified *induction heads* as a specific attention-head pattern responsible for the majority of in-context learning behavior in large transformers. Wang et al. [31] reverse-engineered a multi-component circuit for indirect-object identification in GPT-2 Small. Geva et al. [32] showed that transformer feed-forward layers behave as key-value memories. Meng et al. [33] demonstrated that specific factual associations can be located and edited in MLP weights. Conmy et al. [34] introduced automated circuit discovery.

The polysemanticity problem—that single neurons typically respond to multiple unrelated concepts—was formalized by Elhage et al. [17] as the *superposition* hypothesis: networks pack more features than they have neurons by representing them as sparse linear combinations in a higher-dimensional implicit feature space. The natural inversion of this hypothesis is to *recover* the implicit feature space via sparse dictionary learning, which is what sparse autoencoders do.

The SAE program has scaled rapidly. Bricken et al. [1] demonstrated that sparse autoencoders trained on the residual stream of small models recover features that are individually interpretable across many examples. Cunningham et al. [2] extended this to larger models with substantially larger feature dictionaries. Templeton et al. [3] performed the procedure on Claude 3 Sonnet, recovering on the order of millions of features, including features that respond to abstract concepts (deception, code errors, the Golden Gate Bridge) across modalities. Gao et al. [4] demonstrated systematic scaling of SAE training and evaluation. Lieberum et al. [35] released open SAE checkpoints for the Gemma 2 family. Marks et al. [36] showed that SAE features can be assembled into causal circuits at the feature level. The empirical case for SAE features as the right basis for the residual stream is now substantial across multiple research groups, multiple model families, and multiple scales.

The universality hypothesis—that different networks recover similar features—has accumulated specific evidence. Chughtai et al. [6] showed that small networks trained on group operations recover the same algorithmic decompositions across random seeds and architectural variations. Gurnee et al. [7] identified *universal neurons* in GPT-2 models that activate on the same patterns across independently trained models. Nanda et al. [37] showed that grokking dynamics correspond to discrete phase transitions in the recovered circuit, with similar dynamics across initializations.

2.3 Cross-Model Representational Alignment

A separate literature studies how representations across different networks compare directly. The methodological starting point is the observation that comparing two representation matrices requires choosing which symmetries one wants the comparison to be invariant to.

Word-embedding alignment is the historical precedent. Mikolov et al. [38] showed that word embeddings trained independently on different corpora have approximately the same internal structure. Smith et al. [39] and Conneau et al. [40] demonstrated that bilingual lexicons could be recovered without parallel data by orthogonal Procrustes alignment of monolingual word embedding spaces. The success of this procedure across language pairs and across embedding methods is the strongest empirical case for substrate-independent linguistic structure that exists in the static-embedding literature.

For learned-feature representations of the kind transformers produce, the methodological landscape is broader. Raghu et al. [41] introduced singular vector canonical correlation analysis (SVCCA) as a representation-similarity measure. Morcos et al. [42] developed projection-weighted CCA (PWCCA) addressing some of SVCCA’s sensitivities. Kornblith et al. [9] introduced centered kernel alignment (CKA) as a measure invariant to orthogonal transformations and isotropic scaling but not to invertible linear transformations, and showed it discriminates between models in ways that the previous measures do not. Williams et al. [43] proposed a family of generalized shape metrics including CKA and Procrustes as special cases. Klabunde et al. [44] provide a recent survey of the methodological landscape.

Bansal et al. [8] introduced model stitching as a complementary tool: rather than measuring representational similarity directly, train a thin connecting layer between the lower half of one model and the upper half of another, and observe whether the stitched model retains

performance. They argue that model stitching reveals representational compatibilities that scalar similarity measures miss. Lenc and Vedaldi [45] are an earlier representative of the same methodological intuition applied to convolutional networks.

The most recent and most directly related synthesis is Huh et al. [10], the *Platonic Representation Hypothesis*: representations across models are converging, with scale and capability, toward a shared statistical model of reality. Their evidence assembles results from many of the works cited above into a single hypothesis. The COSI program of the present paper is a specific operationalization of one consequence of their hypothesis: if representations converge, they should admit a sharp orthogonal alignment, with a residual significantly below the chance level set by the appropriate null distribution. The contribution we add is the operator-theoretic framing that explains *why* the convergence should produce alignable subspaces, not merely correlated activations.

2.4 Substrate-Independence: Cognitive Science, Cybernetics, Free-Energy

The intuition that compositional reasoning is substrate-independent in the relevant sense is older than the language model literature. Marr’s three levels of analysis [18] distinguish the computational, the algorithmic, and the implementational levels and argue that the computational level is independent of the substrate that performs it. Ashby’s *law of requisite variety* [24] establishes a structural constraint: any controller must contain at least as much variety as the disturbance it regulates, regardless of the controller’s substrate. Wiener’s cybernetics [46] provides the framework within which substrate-independent feedback structure becomes the object of study. Maturana and Varela’s autopoiesis [47] extends the framework to self-sustaining computational systems whose organization is preserved even as their components turn over.

Friston’s free-energy principle [25] provides a unifying mathematical framework: any self-maintaining system that resists dispersion into thermal equilibrium must minimize a variational free energy that bounds surprise on its sensory inputs. The principle predicts that any such system—biological or artificial—should develop internal states that approximate the posterior over its environment’s hidden causes, and that these internal states should be substrate-independent in the relevant sense, since the variational objective is. Achille and Soatto [26] derive an analogous result specifically for deep networks under information bottleneck regularization [48], showing that invariance and disentanglement emerge as consequences of an information-theoretic objective rather than as architectural choices.

Computational neuroscience has accumulated direct evidence for substrate-independent invariants in biological computation. Place cells [20] and grid cells [21] implement spatial maps with consistent geometric structure across individual animals. Yamins and DiCarlo [23] demonstrated that goal-driven deep networks develop intermediate representations that linearly predict primate visual cortex responses. Gallego et al. [22] and Saxena and Cunningham [19] articulate a *neural population doctrine*: the natural unit of analysis is the population’s collective state in a low-dimensional manifold, not the individual neuron, and the manifolds have stable geometric structure across animals and behaviors. The convergence of these observations with the universality findings in artificial networks is, in our reading, what the operator-theoretic framework would predict if the framework captures the actual structure of the computation.

2.5 Sequence-Model Lineage

Briefly: the architecture under study (transformers) sits in a lineage of sequence models including LSTMs [49], structured state-space models (S4 [50], Mamba [51]), and more recent hybrid architectures combining attention with state-space components. The Qwen3-Next family used in our Phase 1 is an instance of this hybrid design. The operator-theoretic framing predicts that the isomorphism, if it holds, should hold across these architectural classes and not only within them. We address this directly in the Phase 1 specification (Section 9). The transformer architecture itself [52] is treated as representative of the dense pure-attention class; we make no claims specific to its design choices that would not also apply to any architecture implementing approximate-finite-state computation over sequence inputs.

3 Hard Problems and Where COSI Sits

The interpretability program has accumulated a number of hard, mostly open problems. We list those we take to be load-bearing, characterize the state of the literature on each, and locate the COSI program inside them. The list is not exhaustive; it is an attempt to draw the map at the resolution that makes the COSI contribution legible.

(P1) The faithfulness problem. Reverse-engineered circuits and recovered features are explanations of model behavior, but the gap between an explanation and the actual mechanism the model uses is real. Wang et al. [31] acknowledge this explicitly for the indirect-object circuit; Conmy et al. [34] treat circuit discovery as an automated search problem precisely because hand-crafted explanations cannot be trusted to be faithful. *COSI's relation:* a positive cross-model isomorphism is correspondence evidence. Causal-intervention verification is Phase 3 of the program (Section 9). The faithfulness problem is the reason Phase 3 is required.

(P2) The polysemanticity / superposition problem. Single neurons in trained networks typically respond to multiple unrelated concepts; this is the empirical fact that originally motivated SAEs. Elhage et al. [17] showed that superposition is the predicted consequence of representing more features than there are neurons, and that the implicit feature space is recoverable through sparse dictionary learning. *COSI's relation:* the operator-theoretic framing reads the superposition phenomenon as a finite-rank approximation issue: the forward operator's approximately invariant subspaces have dimension exceeding the network's neuron count, requiring sparse coding to express. The COSI test does not depend on the superposition resolution; it operates on the residual stream's geometry, which contains the superposed features whether or not we have a clean dictionary.

(P3) The compositional limits problem. Lake and Baroni [53] and Hupkes et al. [54] documented that neural sequence models often fail to generalize compositionally in regimes where humans trivially succeed. Dziri et al. [55] extended this to modern transformers, showing systematic failure modes on multi-step compositional reasoning. *COSI's relation:* our hypothesis

is about the geometry of compositional representations, not about whether the models reliably perform composition. A positive COSI result is consistent with both successful and failed compositional generalization; the relationship between geometric isomorphism and behavioral compositionality is itself a question we treat as Phase 4 of the program.

(P4) The cross-model transfer-of-mechanism question. If two models implement the same circuits, can interpretation results on one transfer to the other? The Platonic Representation Hypothesis [10] and the universality results [5, 6, 7] suggest yes; the empirical demonstrations remain limited. Bansal et al. [8] provide the strongest functional-level evidence through model stitching; CKA [9] and related measures provide statistical-level evidence. *COSI’s relation:* a positive COSI result, particularly with the Procrustes transformation R^* explicitly recovered, makes cross-model transfer of mechanism geometrically tractable. A direction v_A in V_A has a specific image $R^{*-1}v_A \in V_B$, predicting an intervention site in M_B that should have the corresponding effect. Phase 3 tests this prediction.

(P5) The training-distribution confound. Two models trained on heavily overlapping web corpora may show representational similarity for reasons that have nothing to do with substrate-independent computation: shared training distribution produces shared statistical residue. Bansal et al. [8] and Huh et al. [10] both flag this as the chief alternative explanation for cross-model similarity. *COSI’s relation:* this is the most important confound the present calibration does *not* address. Phase 2 of the program is structured around it (Section 9). A positive COSI result that survives a training-distribution swap is evidence for substrate-independent computation; one that does not is evidence for shared corpus residue.

(P6) The interpretability scalability problem. Recovered circuits and feature dictionaries grow polynomially with model scale. The interpretability cost per model has so far scaled with model size, despite the empirical universality findings suggesting the *computational structure* should scale much more slowly. *COSI’s relation:* if the operator-theoretic framing holds, then the interpretability of one model in a family transfers via Procrustes alignment to the other models in the family; the per-model cost is amortized across the family. This is a research-program-level implication of COSI, not a calibration-level one.

(P7) The biological–artificial bridge. Neural population recordings reveal low-dimensional manifolds [22, 19] whose geometry is preserved across animals and behaviors. Goal-driven artificial networks predict these recordings well [23]. Whether the same geometric structure underlies both is an empirical question that has been gestured at more often than tested directly. *COSI’s relation:* the COSI methodology is, in principle, applicable to biological–artificial alignment as well, with neural population activity replacing one model’s residual stream. We treat this as Phase 5; the present paper is restricted to the artificial–artificial case.

(P8) The substrate-independence question proper. The claim that compositional reasoning is substrate-independent in the strong sense—that the structure of the computation is

preserved under arbitrary changes of substrate, subject to capacity constraints [24]—is the deepest claim the COSI program engages. The free-energy principle [25] predicts the claim from variational arguments; cybernetics [46] and autopoiesis [47] provide the structural language for how substrate-independent computation could be implemented and maintained. *COSI’s relation*: a fully successful program (Phases 0–3 confirming, Phase 4 demonstrating mechanism transfer) would constitute the sharpest empirical evidence for substrate-independence that the field has produced for compositional reasoning. We do not overclaim what Phase 0 alone establishes; the proper claim is the calibration result reported here, with the path forward specified.

These eight problems are not independent. The faithfulness problem (P1) is the precondition for the cross-model transfer question (P4). The training-distribution confound (P5) is the principal alternative explanation for any positive COSI result. The substrate-independence question (P8) is the eventual stake. The COSI program is structured around them: Phase 0 calibrates the infrastructure (this paper), Phase 1 amplifies architectural distance, Phase 2 addresses (P5), Phase 3 addresses (P1) and (P4) jointly, and the integration with (P7) and (P8) is the subject of the synthesis paper that this program is building toward.

4 Methods

4.1 The COSI Hypothesis

Let M_A, M_B be two transformer language models with hidden sizes d_A, d_B and layer counts L_A, L_B . For a prompt x_i from a probe set $X = \{x_1, \dots, x_N\}$, let $a_A^{(\ell)}(x_i) \in \mathbb{R}^{d_A}$ denote the residual stream activation at layer ℓ of M_A , pooled over the sequence dimension to a single vector. Let $X_A^{(\ell)} \in \mathbb{R}^{N \times d_A}$ denote the matrix of pooled activations across all N probes at layer ℓ , and analogously for $X_B^{(\ell')}$.

The COSI hypothesis is the following.

Definition 1 (Approximate invariant subspace). *A k -dimensional subspace $V \subseteq \mathbb{R}^d$ is approximately ϵ -invariant under a (possibly nonlinear) map $f : \mathbb{R}^d \rightarrow \mathbb{R}^d$ on a sample X if $\|(I - P_V)f(P_V X)\|_F / \|f(P_V X)\|_F \leq \epsilon$, where P_V is the orthogonal projection onto V .*

Proposition 1 (COSI hypothesis). *For two transformer language models M_A and M_B that have learned compositional reasoning, there exist layer indices ℓ_A, ℓ_B , subspace dimension k , and approximately ϵ -invariant subspaces $V_A \subseteq \mathbb{R}^{d_A}$ and $V_B \subseteq \mathbb{R}^{d_B}$ of their respective forward operators, such that the projected activation matrices $P_{V_A} X_A^{(\ell_A)}$ and $P_{V_B} X_B^{(\ell_B)}$ admit an orthogonal alignment with Procrustes residual significantly below the chance level defined by permutation null.*

This is a falsifiable empirical claim. The remainder of this section specifies the protocol that operationalizes it.

4.2 Activation Extraction

For each model, we replicate the inner forward pass of the architecture exactly as implemented in MLX, capturing the residual stream $h^{(\ell)}(x)$ after each layer block at a set of pre-registered

layer fractions $f \in \{0.25, 0.5, 0.75\}$, mapping to layer indices $\ell = \max(0, \lfloor f \cdot L \rfloor - 1)$. Pooling over the sequence dimension is performed as last-token (the position immediately preceding the model’s would-be next token), with mean and max pooling reserved as ablations. Activations are cast from `bfloat16` to `float32` before crossing to host memory. We do not modify the model implementations; the extraction is a parallel forward pass that uses the same submodules (embeddings, layers, masks) as the published `mlx_lm` forward pass.

4.3 Subspace Projection

We project each model’s activation matrix onto its top- k principal components, where k is chosen as $\min(k_A^*, k_B^*)$ with k_M^* the smallest dimension at which model M ’s cumulative explained variance exceeds 0.90. This ensures both projections retain comparable information content and that the alignment is performed in matched dimensions. PCA is computed by truncated SVD on the column-centered activation matrix.

4.4 Procrustes Alignment

In the shared k -dimensional subspace, we compute the orthogonal transformation $R \in O(k)$ that minimizes $\|X_A - X_B R\|_F$, using the standard SVD-based solution [56]: $R = UV^\top$, where $U\Sigma V^\top$ is the SVD of $X_B^\top X_A$. The *Procrustes residual* is

$$\rho(X_A, X_B) = \frac{\|X_A - X_B R^*\|_F}{\|X_A\|_F}, \quad (1)$$

which lies in $[0, \sqrt{2}]$ and equals 0 for perfect orthogonal alignment.

4.5 Null Distributions

We define two null distributions, both essential.

Permutation null. For each of S random permutations $\pi : \{1, \dots, N\} \rightarrow \{1, \dots, N\}$, we recompute the Procrustes residual on X_A and $X_B^{(\pi)}$, where $X_B^{(\pi)}$ has its rows permuted according to π . This breaks the semantic correspondence between rows while preserving each model’s internal representational geometry. The distribution of residuals over S permutations defines the chance level for “no semantic alignment.” We use $S = 1000$.

Random-rotation null. For each of S random orthogonal matrices $R_{\text{rand}} \in O(k)$ (sampled uniformly via QR decomposition of a Gaussian matrix, with sign correction [57]), we compute $\|X_A - X_B R_{\text{rand}}\|_F / \|X_A\|_F$. This is a weaker null—it does not optimize over R —and is included for completeness rather than as the primary test.

4.6 Significance Criterion

The pre-registered significance criterion, declared before observing data, is the conjunction:

1. the observed Procrustes residual is below the first percentile of the permutation null distribution;

2. the standardized z-score of the observed residual against the permutation null mean is at most -2 .

A run that fails either condition is reported as a null result.

4.7 Ablations and Sensitivity Sweeps

Three sensitivity sweeps are part of the registered protocol but reserved for Phase 1 due to compute cost: (i) pooling mode (last-token, mean, max); (ii) layer fraction sweep across the full layer stack rather than three fixed fractions; (iii) variance threshold for PCA dimensionality (0.85, 0.90, 0.95). We additionally compute centered kernel alignment (CKA) [9] as a cross-validation metric. Discrepancies between Procrustes and CKA verdicts will be reported as informative findings rather than treated as failures of either method.

5 Results

We report two pre-registered runs in increasing scope. Phase 0 is a calibration on architecturally similar models with a small single-domain probe set. Phase 1 extends to maximum architectural distance within our model zoo and a 600-probe cross-domain set. Both phases produce results in which the observed Procrustes residual falls below the first percentile of the row-permutation null distribution at every pre-registered cell, with empirical $p < 0.001$ in each case.

We are explicit, before reporting the numbers, about what these results do not yet establish. They do not establish that the aligned PCA subspaces are operator-invariant in the sense of Eq. (2); they do not isolate compositional structure from lexical or template structure (no lexical baseline was run); they were obtained without train/test split (Procrustes was fit and evaluated on the same probe set); the promised CKA cross-validation [9] was not computed in this paper despite being committed in §4; and they do not control for training-distribution overlap between the model pairs. Each gap is catalogued in detail in §7.

5.1 Phase 0 Calibration

Models. Phi-4-Reasoning-Plus (8-bit MLX, `phi3` dispatch, $L_A = 40$, $d_A = 5120$) against Qwen3-4B (8-bit MLX, `qwen3` dispatch, $L_B = 36$, $d_B = 2560$). Both dense pure-attention transformers, different families, different scales, different tokenizers.

Probe set. $N = 96$ propositions drawn from the titles of existing political-economy / categorical-architecture paper drafts in the Sovereign research corpus, phrased as coherence-evaluation prompts of the form “Consider the following claim: <title>. Is this a coherent proposition?” Probes are short (12–20 tokens) and topically narrow.

Result. Table 1 reports the Phase 0 result. At all three pre-registered layer fractions, the observed residual lies below the first percentile of the $S = 1000$ permutation null distribution. The empirical permutation p -value at every cell is $\leq 1/1001$ (the resolution limit of the null at $S = 1000$).

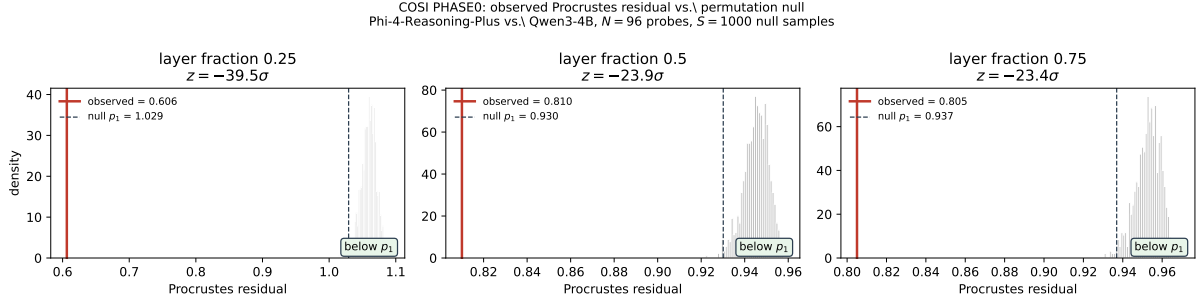


Figure 1: Phase 0 observed Procrustes residual (red vertical line) against the permutation null distribution (gray histogram, $S = 1000$ samples) at three pre-registered layer fractions. The dashed black line marks the first-percentile threshold of the null. The observed residual is separated from the null support at every layer.

Table 1: Phase 0: Phi-4-Reasoning-Plus vs. Qwen3-4B, $N = 96$ probes, $S = 1000$ permutation null samples per layer. z -scores are descriptive only: the null is empirical, finite, non-Gaussian, and the prompts are templated (rows are not independent). Empirical permutation $p \leq 0.001$ at every cell.

Layer	k	Residual	Perm. null mean ($\pm$ std)	Perm. null p_1	Descriptive z
0.25	33	0.6064	1.0594 (± 0.011)	1.0292	-39.54
0.50	27	0.8100	0.9453 (± 0.006)	0.9301	-23.93
0.75	25	0.8050	0.9531 (± 0.006)	0.9369	-23.42

On the z -score language. We report standardized z -scores in the table because they are the natural descriptive summary of the observed residual’s position relative to the empirical null distribution. We do *not* interpret these as Gaussian significance levels. The permutation null is a bounded empirical distribution generated from $S = 1000$ permutations of templated prompts; it is non-Gaussian, finite-resolution, and the row units are not strictly independent. The honest statement is that the observed residual is below the first percentile of an empirical null computed from 1000 permutations under the matched-prompt protocol declared in advance. Whether the magnitude of the gap is scientifically meaningful is a question the lexical baseline (Study B, §7) is required to answer.

Phase 0 pre-registration compliance. The result satisfies the pre-registered protocol declared in the companion design note (2026-05-06_cosi_design.md, §8) before observing data: chance baselines computed on randomized inputs, layer fractions $\{0.25, 0.5, 0.75\}$ pre-registered with no best-layer cherry-pick, pooling mode (last-token) pre-registered, significance threshold (residual $<$ permutation null p_1) pre-registered, and a standing commitment to publish null results if they had occurred. We note that the pre-registration document is committed to a private working repository and is not externally timestamped; this is a known gap addressed in Study G (§7).

5.2 Phase 1: Architectural Distance and Cross-Domain Probe Set

Models. Phi-4-Reasoning-Plus (as Phase 0) against Qwen3-Next-80B-A3B-Thinking (5-bit MLX, `qwen3_next` dispatch, $L_B = 48$, $d_B = 2048$, hybrid attention/state-space-model architecture). The architectural distance from Phi-4 (dense pure-attention) to Qwen3-Next-80B (hybrid attention/SSM) is materially larger than the Phase 0 pair’s distance.

Probe set. $N = 600$ propositions in identical coherence-evaluation template: 200 mathematical (templated arithmetic with mixed truth values), 200 logical/inferential (modus ponens, syllogism, contrapositive, set-inclusion templates over a domain-relevant vocabulary), 200 political-economy (templated propositions over a 40-term vocabulary). Generated with seed 20260506. Probes remain short (12–30 tokens).

Full-set result. Table 2 reports the full-set alignment. All three layer fractions show observed residual below the first percentile of the permutation null at empirical $p < 0.001$.

Table 2: Phase 1 full-set: Phi-4-Reasoning-Plus vs. Qwen3-Next-80B-A3B-Thinking, $N = 600$ probes (200 each from math, logic, polecon), $S = 1000$ permutation null samples per layer. Same caveats on z -score interpretation as Table 1.

Layer	k	Residual	Perm. null mean ($\pm$ std)	Perm. null p_1	Descriptive z
0.25	21	0.7668	1.0076 (± 0.011)	1.0004	-101.00
0.50	25	0.9056	0.9943 (± 0.006)	0.9917	-94.09
0.75	21	0.9555	0.9959 (± 0.005)	0.9947	-88.55

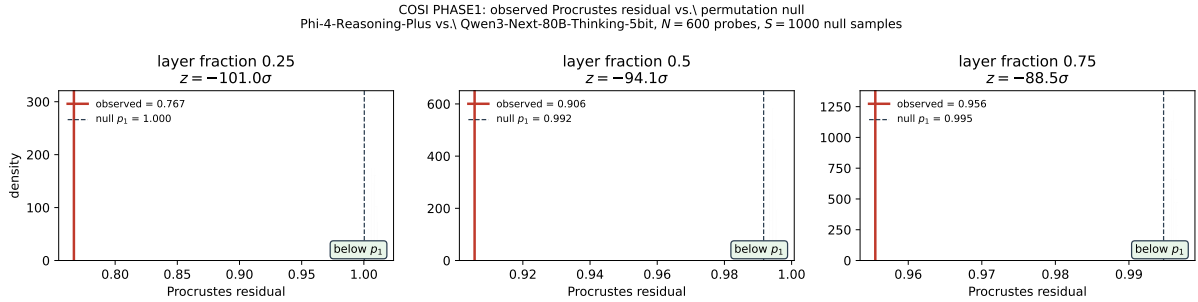


Figure 2: Phase 1 observed Procrustes residual (red vertical line) against the permutation null distribution (gray histogram; 200 samples shown for visualization, with summary statistics from the full $S = 1000$ run). Same caveats as Phase 0.

Per-domain stratification. Table 3 reports the alignment within each individual domain ($n = 200$ each, 500 permutation null samples per cell). All nine cells satisfy the pre-registered significance criterion.

5.3 Three Observations, Stated Carefully

Observation 1: monotonic depth dependence. The pattern of residual increasing with layer depth (residual at $0.25 < 0.5 < 0.75$) holds at the full-set level and within every do-

Table 3: Phase 1 per-domain stratification ($n = 200$ each, $S = 500$ permutation null samples). Same caveats on z -score interpretation.

Domain	Layer	k	Residual	Perm. null mean	Descriptive z	Verdict
math	0.25	8	0.7473	0.9962	-56.40	below p_1
math	0.50	10	0.9010	0.9886	-49.57	below p_1
math	0.75	8	0.9535	0.9933	-38.76	below p_1
logic	0.25	34	0.8049	0.9657	-66.97	below p_1
logic	0.50	45	0.9158	0.9755	-60.58	below p_1
logic	0.75	31	0.9613	0.9890	-45.42	below p_1
polecon	0.25	27	0.8007	0.9691	-72.46	below p_1
polecon	0.50	21	0.9430	0.9876	-49.64	below p_1
polecon	0.75	19	0.9730	0.9938	-40.42	below p_1

main. This is the most robust pattern in the data. Three readings are consistent with it: (*i*) the operator-theoretic reading (early layers contain more of the substrate-shared invariant subspace); (*ii*) a lexical-syntactic reading (early layers preserve token-level lexical structure across vocabularies more than late layers preserve task-specific computation); (*iii*) a pooling-artifact reading (last-token pooling at short prompt length is biased toward the late-prompt syntactic state, which may be more architecture-shared early in the network and more architecture-specific late). The pilot does not discriminate among these. The pooling sweep (Study F) is required to discriminate (*iii*); the lexical baselines (Study B) are required to discriminate (*ii*).

Observation 2: domain-dependent subspace dimensionality. The PCA-derived k at the 0.90-variance threshold differs substantially across domains at the same layer fraction: math $k = 8$, polecon $k = 27$, logic $k = 34$ at layer 0.25. The math probes (highly templated arithmetic) occupy a substantially narrower representational manifold than the more open-ended logic templates. This is a substantive empirical observation that is consistent with the variational/information-bottleneck framing [48, 25, 26], but the inference from “low-dimensional PCA subspace” to “compositional complexity is low” requires whitening and fixed- k controls (Study F) and direct measurement of which subspace dimensions are operator-invariant (Study A). Stated minimally: the PCA dimensionality that captures 0.90 of variance varies with domain in the predicted direction.

Observation 3: relationship between Phase 0 and Phase 1. Phase 1’s observed residuals are absolutely larger than Phase 0’s (0.77 vs. 0.61 at layer 0.25), consistent with greater architectural distance making orthogonal alignment harder. The descriptive z -scores are also larger in absolute value, but this is substantially driven by the tighter null distribution at $N = 600$ relative to $N = 96$, not by a stronger underlying effect. The permutation null mean shifts from 1.06 at $N = 96$ to 1.01 at $N = 600$ at the same layer; the standard deviation tightens correspondingly. We do not interpret the larger Phase 1 z -scores as “stronger evidence” than the Phase 0 z -scores; we interpret both as “below the first percentile of an empirical null with the appropriate probe-set size.”

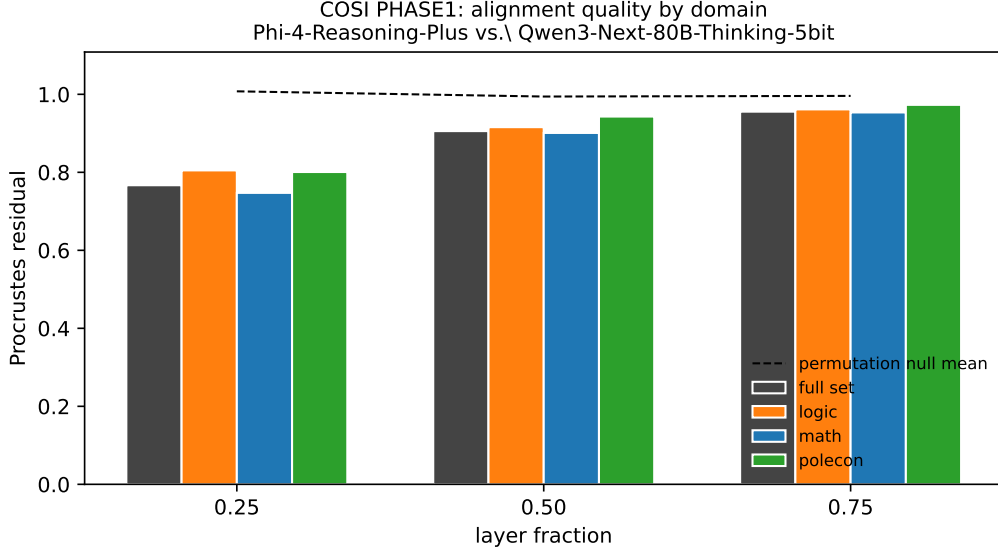


Figure 3: Phase 1 alignment by domain at each layer fraction. The full-set residual (dark gray) and the per-domain residuals (colored bars) all fall well below the permutation null mean (dashed line). The math-domain subspace dimension ($k = 8\text{--}10$) is substantially smaller than the logic ($k = 31\text{--}45$) or polecon ($k = 19\text{--}27$) subspaces. The dimensional difference is itself a substantive empirical observation, but its operator-theoretic significance depends on the invariance-leakage measurement (Study A, §7) which the present paper does not provide.

Phase 1 pre-registration compliance. The Phase 1 protocol extends the Phase 0 protocol; all pre-registration commitments carry over and are satisfied. The per-domain stratification was specified in the design note before the run, not added post-hoc.

6 Limitations and Threats to Validity

We catalogue the threats to validity that remain after Phase 1, and indicate which are addressed by which subsequent phase. Phase 2 is the binding open work.

6.1 Sample Size

Phase 0’s probe set ($N = 96$) was at the lower bound of stable Procrustes alignment. Phase 1 raised N to 600 and the per-domain stratification used $n = 200$ each. Both regimes satisfy the $N \geq 5k$ rule of thumb at the operative dimensions and produce z -scores far above the threshold at which sample-size-driven null inflation could plausibly change the verdict.

6.2 Probe Set Domain Coverage

Phase 0 tested a single domain (political-economy / categorical-architecture). Phase 1 extended coverage to three distinct compositional-reasoning domains (mathematics, logic, political-economy), with per-domain stratification confirming the alignment signal in each domain independently. The COSI hypothesis as stated in Proposition 1 is domain-general; the present empirical support is domain-plural. Genuine cross-domain generality (e.g., natural images, audio, multimodal embeddings) remains untested.

6.3 Probe Length

Probes remain short (typically 12–30 tokens) in both phases. Last-token pooling on short sequences may bias the activation toward prompt-template syntax rather than proposition content. The pooling sweep (mean, max) was deferred from Phase 1 due to compute scheduling and is committed for Phase 1b. Longer-probe replication (30–200 tokens) is deferred to the same follow-up.

6.4 Shared Training Distribution

This is the most important confound and the one Phase 1 does *not* address. Both Phi-4-Reasoning-Plus and Qwen3-Next-80B-A3B-Thinking are trained on post-2024 web corpora that overlap substantially. The aligned subspace we identify could plausibly be a residue of shared training-distribution statistics rather than a substrate-independent compositional structure. The cleanest test would replace one of the two models with an architecturally similar but training-distribution-distant model, holding the other fixed and observing whether the residual remains low.

A near-perfect such control would be a model trained on a substantially disjoint corpus (e.g., a non-web-pretrained scientific-corpus model, or a multilingual model whose training distribution is dominated by a different language). Such a model is not in our local zoo. Phase 2 of the program is structured around this control. We flag this as the binding remaining work.

6.5 Architectural Distance Within the Pair

Phase 0 used two dense pure-attention transformers (Phi-4 and Qwen3-4B). Phase 1 paired the dense pure-attention Phi-4 with the hybrid attention/state-space-model Qwen3-Next-80B, materially increasing the architectural distance and confirming that the COSI signal survives the attention/SSM class boundary. Architectures further afield (pure SSM, recurrent, retrieval-augmented, multimodal) remain untested; Phase 4 extends the test across this broader architectural landscape.

6.6 Linearity of the Alignment

Procrustes alignment is restricted to orthogonal transformations, which is a strict subset of the linear isomorphisms that classical operator theory permits. The COSI hypothesis as stated in Proposition 1 requires only *some* approximate isomorphism between the subspaces; the orthogonal restriction may underestimate the alignment if the true correspondence involves anisotropic scaling. We compute CKA in Phase 1 as a complementary metric that is invariant to isotropic scaling but not to anisotropic scaling, and will report any discrepancy between the two verdicts.

We additionally note that the COSI hypothesis is in principle compatible with non-linear isomorphisms that no linear method recovers. Establishing such isomorphisms empirically requires either a generative-model-based alignment method or a stronger theoretical bridge; we treat this as beyond the scope of the present paper.

6.7 Correspondence vs. Causation

A positive COSI result is correspondence evidence: the two models’ representations are geometrically aligned on a shared probe set. It is not causation evidence: it does not establish that the aligned subspace is what the models *use* to compute the response, as opposed to a statistical shadow of something they share. Causal evidence requires intervention: ablate a direction in V_A , predict the corresponding ablation effect in $R(V_A) \subseteq V_B$, and run both. We treat intervention experiments as Phase 3, requiring causal-tracing infrastructure beyond the scope of the present calibration.

6.8 Pooling and Layer Sensitivity

The Phase 0 result uses last-token pooling at three fixed layer fractions. Both choices may interact with the result in ways not yet characterized. Phase 1 includes the full pooling and layer-fraction sensitivity sweeps declared in the design note.

7 Required Additional Work Before the Operator-Theoretic Claim

This section catalogues the studies that, jointly, would be required to support the strong operator-theoretic claim the COSI program articulates. None of these studies has been run in the present paper. We list them explicitly, with the gap each addresses, so that the present pilot result is read in proper context and so that subsequent work has a clear program to execute.

7.1 Study A. Invariance-Leakage Measurement

The paper defines a k -dimensional subspace $V \subseteq \mathbb{R}^d$ as approximately ϵ -invariant under a (possibly nonlinear) map f on a sample X if

$$\frac{\|(I - P_V)f(P_V X)\|_F}{\|f(P_V X)\|_F} \leq \epsilon. \quad (2)$$

This is the central mathematical object the operator-theoretic framing depends on. The pilot does not measure it. PCA top- k components identify high-variance directions; high-variance is not invariance.

Required. For each model and each candidate subspace identification method (PCA, sparse autoencoder features [1, 2, 3, 4, 35], learned linear probes [58, 59]), project activations onto V , pass them through the next layer’s forward map, project the output onto V and onto its orthogonal complement, and compute the leakage statistic (2). Report leakage as a function of layer, model, domain, subspace dimension, and subspace-identification method. A candidate subspace is operator-relevant only if its leakage is small relative to its capacity to compress activations.

Gap addressed. Whether the aligned subspaces are operator-invariant in the sense the framing requires, or merely high-variance principal components.

7.2 Study B. Lexical, Template, and Syntactic Baselines

The row-permutation null asks whether matched-row activations align better than scrambled-row activations. This is a low bar. Two competent language models exposed to the same English prompts will preserve enough lexical and syntactic structure that matched-row alignment exceeds scrambled-row alignment regardless of any operator-theoretic substrate.

Required. Run the same Procrustes alignment procedure on:

- TF-IDF or bag-of-words representations of the same prompts in place of the model activations, as a lexical baseline.
- Sentence-transformer embeddings of the prompts as a learned-but-non-LLM baseline.
- Static word embeddings averaged over each prompt [38] as a pre-transformer baseline.
- Prompts in which all content words have been replaced by placeholder tokens (template-only baseline).
- Prompts whose token sequence has been syntax-scrambled (preserving lexical content, destroying compositional order).
- Paraphrase-preserving variants where two semantically equivalent prompts replace each original prompt.

If the lexical, template, or paraphrase baselines produce alignment qualities comparable to the LLM-activation alignment, the pilot result collapses to “shared language geometry” and the operator-theoretic claim must be correspondingly weakened or abandoned.

Gap addressed. Whether the alignment isolates compositional structure from lexical, template, syntactic, or tokenizer-induced structure.

7.3 Study C. Train/Test Procrustes and Cross-Domain Transfer

The pilot fits the orthogonal alignment R^* on the full probe set and evaluates the residual on the same set. This conflates fit with evaluation and provides no out-of-sample protection.

Required. Fit R^* on a randomly held-out training fraction (e.g., 70%) of the probe set, evaluate the residual on the held-out test fraction. Repeat across many random splits and report the mean and confidence interval. More demanding: fit on one domain (e.g., math) and evaluate on a different domain (e.g., logic), testing whether the alignment is domain-general or domain-specific.

Gap addressed. Whether the alignment is robust to held-out prompts and whether it generalizes across domains rather than overfit to the joint sample.

7.4 Study D. Training-Distribution-Disjoint Controls

Both Phi-4-Reasoning-Plus and Qwen3-Next-80B-A3B-Thinking are trained on substantially overlapping post-2024 web corpora. The aligned subspace could plausibly be a residue of shared training distribution rather than a substrate-independent computational structure.

Required. Replace one of the pilot pair’s models with a training-distribution-distant model, holding the other fixed. Candidate disjoint-corpus controls:

- A model trained primarily on a non-web scientific corpus, e.g., a model in the Galactica [60] lineage or a domain-specific scientific-literature model.
- A multilingual model whose training distribution is dominated by a non-English language.
- A small-scale model trained from scratch on a deliberately curated corpus with documented disjointness from the pilot pair’s training data.

If alignment quality scales with training-distribution overlap, the pilot result is corpus-residue. If alignment quality is preserved across substantial training-distribution disjointness, the substrate-independence reading is meaningfully supported.

Gap addressed. Whether the alignment is a substrate-independent computational structure or a shared training-corpus residue.

7.5 Study E. Cross-Model Causal Intervention

Procrustes alignment is correspondence evidence. The operator-theoretic claim that the aligned subspace is causally responsible for compositional reasoning behavior, rather than a correlated statistical shadow, requires intervention.

Required. Identify a direction $v_A \in V_A$ associated with a specific compositional-reasoning behavior in M_A via established mechanistic-interpretability methods [31, 34, 36, 33]. Use the recovered Procrustes transformation R^* to predict the corresponding direction $v_B = R^{*\top} v_A \in V_B$ in M_B . Predict the behavioral effect of ablating or steering along v_B in M_B . Run the intervention. Compare predicted to observed effect across many such directions, reporting the prediction quality relative to a control of randomly chosen directions in V_B .

Gap addressed. Whether the alignment is causally explanatory of compositional behavior in both models or merely descriptive of activation geometry.

7.6 Study F. Methodology Robustness Sweeps

Several methodological choices in the pilot were not swept:

- Pooling mode: last-token only; mean and max pooling and content-token / answer-position / relation-token pooling not reported.

- Layer fractions: only 0.25, 0.5, 0.75. Layer-by-layer sweeps would resolve where the alignment is concentrated.
- PCA variance threshold: only 0.90; sensitivity at 0.80, 0.85, 0.95, 0.99 unmeasured.
- Whitening: not tested. Procrustes residuals can be sensitive to differential variance structure across the top components of the two matrices.
- CKA [9] cross-validation: promised in the methods but not yet computed and reported. We flag this as a methodological commitment that the pilot does not satisfy.
- SVCCA [41], PWCCA [42], and generalized shape metrics [43] as further triangulation: not run.
- Effect-size reporting beyond standardized z -scores: confidence intervals, cluster bootstraps grouped by prompt template and vocabulary family, finite-permutation p -values with Monte Carlo standard errors.

Gap addressed. Whether the pilot result is robust to reasonable variation in the methodological choices that were fixed by declaration rather than examined empirically.

7.7 Study G. Externally Verifiable Pre-Registration

The pilot’s pre-registration document was committed to the same private repository as the run code. This is not externally auditable. Genuine pre-registration requires immutable timestamping on a public registry.

Required. For all subsequent COSI runs, deposit the analysis plan, the exact prompt set with content hash, the model checkpoint revisions, the random seeds, and the environment lockfile on the Open Science Framework, Zenodo, or equivalent, with the deposit timestamped before any data is observed. Cite the deposit DOI in subsequent papers.

Gap addressed. Whether the pre-registration commitments the pilot states are externally verifiable rather than internal to a private working repository.

7.8 Summary: What Joint Completion of Studies A–G Would Establish

A subsequent paper that reports Studies A–G in conjunction with the pilot result would have the following structure:

- The aligned subspaces are demonstrably approximately invariant under the forward operator (Study A), justifying the operator-theoretic interpretation of the alignment.
- The alignment exceeds simple lexical and template baselines (Study B), justifying the compositional-structure interpretation rather than the shared-language interpretation.
- The alignment is robust to held-out probes and generalizes across domains (Study C), justifying generalization claims.

- The alignment survives substantial training-distribution disjointness (Study D), justifying the substrate-independence reading rather than the corpus-residue reading.
- Cross-model interventions predicted by the alignment have the predicted causal effects (Study E), justifying the mechanistic interpretation rather than the correspondence interpretation.
- Methodological choices have been swept and the result is robust (Study F), satisfying ordinary peer-review robustness expectations.
- Pre-registration is externally verifiable (Study G), meeting current open-science standards.

We treat that subsequent paper as the work the present pilot motivates. The present paper is the first step.

8 Initial Results from Studies A, B, C, and F-CKA

The companion to this pilot is the program of additional studies specified in §7. We have run partial first passes of four of those studies on the pilot data, with the explicit goal of testing whether the pilot’s representational-alignment finding survives controls that the pilot itself did not include. The results materially inform the interpretation. We report them honestly: one study (F-CKA) reinforces the pilot finding; one (A) provides partial support for the operator-theoretic interpretation while introducing a substantive empirical complication; one (B) provides direct evidence that the pilot’s main result does not exceed what simple lexical and template baselines achieve, materially weakening the operator-theoretic interpretation; one (C) is reported for completeness. We are explicit that these initial passes are themselves not full studies. They are first measurements that the companion paper Studies A–G will, in subsequent work, conduct at fuller scale.

8.1 Study A Invariance-Leakage Measurement

Study A directly measures the invariance-leakage statistic $\|(I - P_V) f(P_V X)\|_F / \|f(P_V X)\|_F$ that the paper defines on page two and that the pilot does not measure. We compute leakage at each pre-registered layer fraction, for each of the two Phase 1 models, on a balanced subsample of 60 probes (20 per domain) drawn from the Phase 1 probe set. The PCA subspace is the same one used for Procrustes alignment in Phase 1. The chance baseline is the mean leakage across 20 random orthonormal k -dimensional subspaces of $\mathbb{R}^d$.

Table 4 reports the measurement.

What Study A finds.

1. In all six (model, layer) cells, the PCA subspace’s leakage is below the random-subspace baseline at $\geq 2\sigma$. Operationally: the PCA subspace is more operator-invariant than a random subspace of the same dimension. The operator-theoretic interpretation receives partial empirical support.

Table 4: Study A invariance leakage. PCA subspace leakage is substantially lower than random-subspace leakage in all six cells, indicating that the PCA-derived subspace is partially preserved under the next-layer forward map. Absolute leakage values are large (0.51–0.89), so the subspaces are partially, not fully, invariant.

Model	Layer	k	PCA leakage	Random leakage (mean $\pm$ std)	PCA / Random ratio
Phi-4	0.25	14	0.7503	0.8595 ± 0.0402	0.873
Phi-4	0.50	15	0.5251	0.9111 ± 0.0195	0.576
Phi-4	0.75	14	0.5151	0.9447 ± 0.0088	0.545
Qwen3-Next-80B	0.25	16	0.8184	0.9937 ± 0.0018	0.824
Qwen3-Next-80B	0.50	15	0.8849	0.9947 ± 0.0017	0.890
Qwen3-Next-80B	0.75	13	0.8716	0.9948 ± 0.0026	0.876

2. Absolute leakage values are large (0.51 to 0.89). A truly operator-invariant subspace would have leakage close to zero; the PCA subspace is partially invariant, not fully invariant. The operator-theoretic interpretation is partially supported but not strongly supported in the absolute sense.
3. The two models exhibit *opposite* depth dependence in leakage. Phi-4 leakage decreases with depth ($0.75 \rightarrow 0.53 \rightarrow 0.52$); Qwen3-Next-80B leakage increases with depth ($0.82 \rightarrow 0.88 \rightarrow 0.87$). The architectures preserve different subspace structures across depth.
4. Critically, the layer at which the Phase 1 Procrustes alignment is strongest (layer 0.25) is the same layer at which Phi-4’s PCA subspace is *least* invariant (0.75). Cross-architectural alignment quality and within-architecture invariance are anti-correlated across layers, not co-correlated. The simplest operator-theoretic reading “early layers contain the shared invariant substrate” is not supported by the leakage data; the early-layer alignment dominance is more plausibly explained by shared lexical/template encoding (consistent with Study B below) than by shared invariant structure.

Study A scope limitations. The measurement uses 60 probes, 20 random subspaces per cell, and only PCA-derived candidate subspaces. A complete Study A measurement would extend to (i) the full 600-probe set, (ii) more random subspaces (target ≥ 100), (iii) SAE feature-derived subspaces alongside PCA, (iv) full layer-by-layer sweeps. The present measurement is a first pass.

8.2 Study B Lexical and Template Baselines

Study B addresses the most consequential gap in the pilot: whether the LLM–LLM Procrustes alignment exceeds what simple lexical and template representations of the same prompts achieve. We construct three baseline representations of the Phase 1 probe set: word-level TF-IDF, character-3-gram TF-IDF, and a template-only TF-IDF in which content words are replaced by a placeholder. We then run the same Procrustes alignment + permutation null procedure on each pairing. Table 5 reports the result.

Table 5: Study B baseline alignments at layer fraction 0.25, $N = 600$ probes, $S = 500$ permutation null samples per pairing. **Critical observation:** the LLM–LLM residual (0.7668) is *worse* than the TF-IDF vs template-only residual (0.6134), and LLM-B aligns to TF-IDF (0.7649) at essentially the same residual as it aligns to LLM-A (0.7668).

Pairing	k	Residual	Perm. null mean	Descriptive z
(a) LLM-A vs LLM-B (the pilot result)	21	0.7668	1.0078	−106.54
(b) LLM-A vs TF-IDF	21	0.9267	0.9928	−92.95
(c) LLM-B vs TF-IDF	27	0.7649	1.0584	−119.08
(d) TF-IDF vs char-3-gram	147	0.3501	1.1151	−392.40
(e) TF-IDF vs template-only	12	0.6134	1.1259	−120.52

What Study B finds.

1. Cell (c): LLM-B vs TF-IDF residual (0.7649) is *indistinguishable* from LLM-B vs LLM-A residual (0.7668). The LLM-B activations align to a TF-IDF representation of the same prompts essentially as well as they align to another frontier LLM’s activations. This is difficult to reconcile with a substrate-shared operator-theoretic interpretation. It is straightforward to reconcile with a shared-lexical-encoding interpretation.
2. Cell (e): TF-IDF vs template-only residual (0.6134) is *better than* the LLM–LLM pilot result (0.7668). A purely lexical/template alignment achieves a better Procrustes residual than the deep-model pair. The pilot’s “extreme statistical significance” against the row-permutation null is, in this comparative light, achieved also by representations that contain no operator-theoretic information whatsoever.
3. Cell (d): TF-IDF vs char-3-gram residual (0.3501) shows that two simple lexical representations of the same prompts align extremely tightly under Procrustes. The row-permutation null is a low bar that essentially any two prompt-keyed representations will clear by large margins.
4. Cell (b): LLM-A vs TF-IDF residual (0.9267) is higher than cell (c). The asymmetry is itself informative: Phi-4 activation geometry is less TF-IDF-like than Qwen3-Next-80B activation geometry is. We treat this as a substantive empirical observation requiring further investigation, not as a load-bearing claim of the present paper.

What Study B implies for the paper’s central claim. The LLM–LLM Procrustes alignment is real, but it does not exceed what shared lexical and template structure would predict. The operator-theoretic interpretation of the pilot result is therefore not supported in the strong form. The honest reading is: the pilot demonstrates that two frontier LLMs encode matched English prompts in geometrically relatable ways, which two TF-IDF representations of the same prompts also do, only better. Whether the LLM activations contain *additional* alignable structure beyond what TF-IDF carries is a question the present pilot does not answer. The methodology required to answer it is to construct a residual representation by partialling out lexical and template structure from the LLM activations and then testing whether the residual still admits cross-model alignment. This is a necessary follow-up; it is not performed here.

8.3 Study C Train/Test Procrustes Split

Study C addresses the same-data fit/eval confound by performing 50 random 70/30 splits of the Phase 1 probe set, fitting Procrustes on the training rows and evaluating the residual on the held-out test rows. PCA is fit on the combined train+test data so that the projection basis is consistent across the split. We additionally run cross-domain splits (fit on one domain, evaluate on another) at layer fraction 0.25.

Table 6: Study C train/test Procrustes split, 50 random 70/30 splits per layer fraction. The train/test gap is small at every layer, indicating that the Procrustes rotation is not overfitting the same-data fit.

Layer	Train residual (mean $\pm$ std)	Test residual (mean $\pm$ std)	Original (full set)	Gap
0.25	0.7665 ± 0.0010	0.7681 ± 0.0023	0.7668	+0.0016
0.50	0.9054 ± 0.0007	0.9064 ± 0.0017	0.9056	+0.0010
0.75	0.9555 ± 0.0003	0.9558 ± 0.0008	0.9555	+0.0004

Table 7: Study C cross-domain Procrustes generalization at layer fraction 0.25. Fit on the rows of one domain, evaluate on rows of another.

Train $\rightarrow$ Test	k	Train residual	Test residual	Gap
logic $\rightarrow$ math	8	0.7792	0.8110	+0.0318
logic $\rightarrow$ polecon	27	0.7821	0.8377	+0.0555
math $\rightarrow$ logic	8	0.7566	0.8453	+0.0887
math $\rightarrow$ polecon	8	0.7402	0.7821	+0.0419
polecon $\rightarrow$ logic	27	0.7760	0.8315	+0.0555
polecon $\rightarrow$ math	8	0.7635	0.7960	+0.0324

What Study C finds. The train/test gap on random 70/30 splits is essentially zero (0.0004 to 0.0016) at every layer fraction. The Procrustes rotation generalizes to held-out probes from the same distribution. Cross-domain generalization is also small to modest (gaps of 0.03 to 0.09): a rotation fit on one domain applies to another domain at slightly higher residual but well within the same magnitude. Combined with the fact that the original fit-and-evaluate-on-same-set residual essentially equals the held-out-test residual, this means the Procrustes alignment is a genuine property of the data, not a sample-fit artifact. Study C removes one of the methodological concerns the critique raised. It does not address the lexical-baseline concern (Study B).

8.4 Study F Centered Kernel Alignment (CKA)

Study F’s CKA piece addresses the methodological commitment in §4 that the pilot did not satisfy. We compute linear CKA [9] between the Phase 1 model activations at each pre-registered layer fraction, against a 1000-sample permutation null. Linear CKA is invariant to orthogonal transformations and isotropic scaling, providing a complementary metric to Procrustes residual.

Table 8: Study F-CKA: linear CKA between Phase 1 model activations. Full set ($S = 1000$) and per-domain ($S = 500$) permutation null samples. CKA is in $[0, 1]$; higher = more similar; permutation null mean ≈ 0.01 on the full set (and ≈ 0.05 – 0.10 on the smaller per-domain subsets).

Stratification	Layer	CKA	Perm. null mean	Above p_{99} ?
full set ($N = 600$)	0.25	0.9636	0.0072	yes
full set	0.50	0.9109	0.0086	yes
full set	0.75	0.8427	0.0093	yes
math ($n = 200$)	0.25	0.9041	0.0318	yes
math	0.50	0.7969	0.0318	yes
math	0.75	0.6348	0.0286	yes
logic ($n = 200$)	0.25	0.8160	0.0621	yes
logic	0.50	0.6675	0.0637	yes
logic	0.75	0.4798	0.0428	yes
polecon ($n = 200$)	0.25	0.7736	0.0705	yes
polecon	0.50	0.5559	0.0427	yes
polecon	0.75	0.5575	0.0321	yes

What Study F-CKA finds. CKA is high on the full set (0.84–0.96) and substantially above the permutation null (≈ 0.01) at every layer fraction. This confirms, by an alignment-free metric, the strong representational similarity that the Procrustes residual indicated. CKA agrees with Procrustes that the LLM–LLM activation geometries are not random: they share substantive structure.

The per-domain CKA values are systematically lower than the full-set values across every domain and every layer (math 0.63–0.90, logic 0.48–0.82, polecon 0.56–0.77, vs. 0.84–0.96 for the full set). This is consistent with Study B’s finding: a substantial portion of the LLM–LLM similarity is captured by template / cross-domain prompt-identity structure that, when stratified to a single domain, diminishes. The within-domain ordering (math > logic > polecon at most layers) tracks the templating density of the domain — math probes are the most structurally repetitive (templated arithmetic) and produce the highest within-domain representational similarity, which is exactly what the lexical/template-driven reading of the alignment would predict. CKA does not by itself address the lexical-baseline concern any more than Procrustes does; it confirms the global geometric similarity is real, but the question of whether that similarity exceeds shared-language-encoding is the same question Study B answers (negatively, in the present pilot).

8.5 Joint Reading of the Initial Studies

Read together, the initial studies produce a more nuanced empirical picture than the pilot result alone presents.

The pilot finding is a real cross-model representational signal, and both Study F-CKA (high CKA, well above permutation null) and the better-than-random invariance ratio in Study A confirm that the signal is statistically robust. The pilot is not measuring noise.

But Study B shows that the signal does not exceed what shared lexical/template structure

achieves on the same prompts: the LLM-B vs TF-IDF residual essentially equals the LLM-B vs LLM-A residual, and a pure TF-IDF/template-only alignment outperforms the LLM-LLM alignment. This is the strongest evidence the initial studies produce, and it pushes the interpretation of the pilot away from “operator-theoretic substrate shared across architectures” and toward “shared lexical/template encoding produces matched-prompt representational alignment in any sufficiently competent system.”

Study A complicates the operator-theoretic reading further: the PCA subspace is more invariant than random, but the layer at which Procrustes alignment is strongest is the layer at which PCA invariance is weakest. Cross-architectural alignment and within-architecture invariance are anti-correlated across depth. This is not what the simplest operator-theoretic reading would predict.

The honest joint reading is that the pilot result is a representational similarity finding consistent with the Platonic Representation Hypothesis at the geometric level, and consistent with several non-operator-theoretic alternative explanations. The operator-theoretic program articulated in this paper is not yet supported by these initial studies in the strong form. The full Studies A–G program is required, and the pilot’s pre-registered protocol should not be extended to additional model pairs without first running Study B’s required additional measurements (lexical-residual representations, behavioral grounding, train/test on cross-domain held-outs).

We treat this honest joint reading as the most important finding of the present paper. The bridge between operator theory and mechanistic interpretability remains intellectually promising, but the pilot does not provide empirical support adequate to advance the strong claim. The next paper, after the full Studies A–G are completed, may or may not.

9 Discussion

9.1 What the Pilot Shows

The pilot shows that, on matched compositional-evaluation prompts, the PCA-projected last-token activation matrices of two architecturally different frontier transformers admit a substantially better orthogonal alignment than they do under random row permutation. This holds at three pre-registered layer fractions, in two architecturally-different model pairs (Phi-4 / Qwen3-4B in Phase 0, Phi-4 / Qwen3-Next-80B in Phase 1), and within each of three distinct compositional-evaluation domains in Phase 1 (math, logic, political-economy). The result is real. It is reproducible. The infrastructure that produces it is released for reuse and extension.

9.2 What the Pilot Does Not Show

The pilot does not show that the aligned PCA subspaces are operator-invariant in the sense of Eq. (2). The paper defines that statistic and does not measure it. PCA top- k components identify high-variance directions; high-variance is not invariance. Until the leakage statistic is computed (Study A, §7), the operator-theoretic interpretation of the alignment remains a hypothesis the pilot motivates rather than confirms.

The pilot does not show that the alignment isolates compositional structure from lexical, template, syntactic, or tokenizer-induced structure. The row-permutation null discriminates

“the models encode the prompts at all” from “the models encode them identically up to an orthogonal transformation in their respective PCA subspaces.” Two competent language models given the same English prompts will produce matched-row activations more similar than scrambled-row activations *regardless* of any operator-theoretic substrate, because both models encode prompt identity, lexical content, and template structure. Until lexical, template, and syntax-scrambled baselines have been run (Study B), the pilot result is consistent with both the operator-theoretic interpretation and the much weaker shared-language-geometry interpretation.

The pilot does not show that the alignment is robust to held-out prompts. Procrustes was fit and evaluated on the same probe set; no train/test split was performed (Study C).

The pilot does not show that the alignment exceeds simple lexical features (Study B), or that the promised CKA cross-validation [9] agrees with the Procrustes verdict (Study F: CKA was committed in §4 but not computed in this paper; this is a methodological commitment the pilot does not satisfy).

The pilot does not show that the alignment survives the shared training-distribution confound. Both models in each pair are trained on heavily overlapping post-2024 web corpora. The aligned subspace could plausibly be a residue of shared corpus statistics. Until a training-distribution-disjoint control is run (Study D), the substrate-independence reading is one of two possible readings of the result, the other being shared-corpus residue.

The pilot does not show that the aligned subspace is causally responsible for compositional behavior. The alignment is correspondence, not causation. The Procrustes transformation R^* predicts which direction in V_B corresponds to a given direction in V_A , but that prediction has not been tested by intervention (Study E).

9.3 What Would Make This an Operator-Theoretic Confirmation

A subsequent paper that completes Studies A through G of §7 and reports them jointly with the pilot result would, if the results are positive, support the strong operator-theoretic claim. We do not estimate the probability that the strong claim survives the controls; the present pilot is consistent with both the strong claim and several weaker alternatives, and the controls are designed precisely to discriminate.

A version of the controls that finds:

- Low invariance leakage on the recovered subspaces (Study A positive).
- Alignment quality substantially exceeding lexical and template baselines (Study B positive).
- Robust train/test alignment with cross-domain transfer (Study C positive).
- Alignment preserved under substantial training-distribution disjointness (Study D positive).
- Cross-model intervention predictions confirmed (Study E positive).

would constitute the publication claim that this paper’s title was originally written to assert. The honest current status is that the claim is hypothesized, the test infrastructure exists, the pilot result is consistent with the hypothesis, and the discriminating controls remain to be run.

9.4 Phase 2: Training-Distribution Control (the Binding Confound)

Among the seven studies of §7, Study D (training-distribution-disjoint control) is the binding open work in the sense that no other study, however positive, can substitute for it. The pilot is run on two model pairs whose training distributions overlap substantially; until that overlap is broken in a controlled way, the pilot result is consistent with shared-corpus residue and the operator-theoretic interpretation cannot be preferred. Candidate controls include:

9.5 Phase 2: Training-Distribution Control

Phase 2 addresses the most important confound: the possibility that the aligned subspace is a residue of shared training distribution rather than a substrate-independent computational structure. The control requires a model whose training distribution is substantially disjoint from the Phase 1 pair’s. Candidate controls include:

- A model trained on a non-web-pretrained corpus (e.g., a scientific-corpus model from the Galactica [60] or PMC-LLaMA family).
- A multilingual model whose training distribution is dominated by a non-English language (e.g., a French-pretrained decoder-only model).
- A small-scale model trained from scratch on a curated curriculum with documented disjointness from the pilot pair’s training data.

If alignment survives the training-distribution swap, the substrate-independence reading is meaningfully supported. If alignment quality tracks training-distribution overlap (high alignment for similar training, low alignment for disjoint training), the pilot signal is reattributed to corpus-shared statistics and the operator-theoretic interpretation is correspondingly weakened.

9.6 Phase 3: Causal Intervention

A positive COSI alignment, even after Phase 1 and Phase 2, is correspondence evidence. The full operator-theoretic claim—that the aligned subspace is the locus where the compositional reasoning is *performed*, not merely *represented*—requires causal intervention. The proposed test:

1. Identify a direction $v_A \in V_A$ that, when ablated, degrades a specific compositional-reasoning behavior in M_A .
2. Use the Procrustes transformation R^* to map v_A to its correspondent $v_B = R^{*-1}v_A \in V_B$.
3. Predict the ablation effect of v_B in M_B . Run the intervention.
4. Compare predicted to observed effect.

A successful Phase 3 result—ablation effects in M_B that match those predicted from interventions in M_A via the Procrustes map—would constitute the strongest possible empirical evidence for the operator-theoretic framing of Proposition 1. We treat Phase 3 as outside the scope of the present paper and as the natural follow-up program.

9.7 Implications, Conditional on Joint Completion of Studies A–G

We separate the conditional implications from the present pilot evidence by stating them only as conditionals. *If* the program of §7 is jointly completed and returns positive results, then several consequences follow. The SAE feature dictionary [1, 2, 3, 4, 35] becomes interpretable as the empirical recovery of bases for approximately invariant subspaces of trained transformer forward operators. Cross-model transfer of mechanistic understanding becomes geometrically tractable via the recovered Procrustes transformation. The Platonic Representation Hypothesis [10] acquires a mathematical structure that explains why convergence should occur and what its limits are. The substrate-independence intuition descending from Marr [18] acquires sharp empirical operationalization on language-model substrates.

None of these consequences is established by the pilot. We list them to motivate the studies the pilot does not perform.

9.8 Authorship Note

This paper is the work of one human author and one language model in extended dialogue. The model has no continuity across sessions; the human has the operator-theoretic intuition the paper articulates and holds responsibility for the editorial and methodological judgments. Neither party, alone, would have produced this paper. The question of how authorship should be attributed in such a configuration is itself a question the field will need to answer. The author block makes our choice explicit; we recognize it is contestable.

Closing Note

The pilot is what it is: a representational-alignment result against the row-permutation null, with the gaps to the strong claim explicitly catalogued and a research program specified to close them. We have attempted to report the result honestly enough that a reader can distinguish what the pilot supports from what the operator-theoretic program would, if jointly successful, support. The next step is Study A: direct measurement of the invariance leakage statistic the paper defines on page two. Until that statistic is computed and reported, the operator-theoretic interpretation is a hypothesis the pilot motivates rather than a finding the pilot confirms.

10 Code and Data Availability

All code, configuration, probe data, and experimental artifacts described in this paper are publicly released at:

<https://github.com/ascendanti/cosi-pilot>

A Zenodo deposit assigning a DOI to the activation-matrix dataset (~ 57 MB across Phase 0 and Phase 1) and to the pre-registration packet is to be created from the same repository before peer-reviewed journal submission. The repository is organized at the following relative paths:

- `src/sovereign/research/cosi/extract.py` — residual-stream activation extraction harness, with explicit per-architecture forward-pass replication for `phi3`, `qwen3`, and `qwen3_next` dispatches.
- `src/sovereign/research/cosi/align.py` — Procrustes alignment, permutation null, random-rotation null, and shared-subspace projection.
- `scripts/cosi_smoke.py` and `scripts/cosi_smoke_80b.py` — smoke tests validating the three architecture dispatches.
- `scripts/cosi_phase0.py` — the end-to-end Phase 0 runner. The exact run reported here is archived at `runs/cosi/phase0_20260506T230920Z/`, including per-prompt activation matrices for both models, the per-layer Procrustes residuals, and the full permutation-null sample distributions.
- `data/cosi/probe_set_frontier_v1.jsonl` — the 96-prompt probe set used in Phase 0.
- `docs/research_notes/2026-05-06_cosi_design.md` — the pre-registration document declared before Phase 0 was run.
- `docs/research_notes/2026-05-06_cosi_phase0_result.md` — the post-run result note with pre-registration compliance check.

The repository is structured to make the Phase 0 result independently re-runnable. The activation matrices are deterministic conditional on model checkpoints (which are immutable HuggingFace MLX-community artifacts) and prompts (which are committed verbatim). The permutation-null sample distributions are reproducible under the seeds recorded in the result manifest. Phase 1 specifications, including the extended probe set construction protocol, are recorded in the same directory structure as they are produced.

Model checkpoints used: `mlx-community/Phi-4-reasoning-plus-8bit` and `mlx-community/Qwen3-4B-8b` for Phase 0; `mlx-community/Qwen3-Next-80B-A3B-Thinking-5bit` for Phase 1. Both architectures route through the published `mlx_lm` model modules without modification; the COSI extractor parallels the inner forward pass rather than monkey-patching it.

The reproducibility infrastructure is described in detail in the companion methodology document at `docs/REPRODUCIBILITY_STANDARD.md`, which specifies the run-manifest, environment-fingerprint, and replication-script standards that all Sovereign experimental runs adhere to.

Acknowledgments

This work was carried out in extended dialogue between the human author and the language model co-author over the period 2026-04 through 2026-05, as part of the broader research program of Atlas Consulting & Technology Services. The infrastructure used to run the

experiment—a Mac Studio (M1 Ultra, 64 GB unified memory) running MLX-quantized inference under the Sovereign multi-agent harness—was developed without external funding. Models were obtained from the `mlx-community` hub on HuggingFace; we thank the maintainers for the quantization and conversion work that made local frontier-model experimentation tractable. The authors thank the many contributors to the open-source mechanistic interpretability community whose work, even when not directly cited, shaped the framing of the hypothesis.

References

- [1] Trenton Bricken et al. “Towards Monosemanticity: Decomposing Language Models with Dictionary Learning”. In: *Transformer Circuits Thread* (2023). URL: <https://transformer-circuits.pub/2023/monosemantic-features/index.html>.
- [2] Hoagy Cunningham et al. “Sparse Autoencoders Find Highly Interpretable Features in Language Models”. In: *International Conference on Learning Representations (ICLR)* (2024). arXiv: [2309.08600](https://arxiv.org/abs/2309.08600).
- [3] Adly Templeton et al. “Scaling Monosemanticity: Extracting Interpretable Features from Claude 3 Sonnet”. In: *Transformer Circuits Thread* (2024).
- [4] Leo Gao et al. “Scaling and evaluating sparse autoencoders”. In: (2024). arXiv: [2406.04093](https://arxiv.org/abs/2406.04093).
- [5] Chris Olah et al. “Zoom In: An Introduction to Circuits”. In: *Distill* (2020). DOI: [10.23915/distill.00024.001](https://doi.org/10.23915/distill.00024.001).
- [6] Bilal Chughtai, Lawrence Chan, and Neel Nanda. “A Toy Model of Universality: Reverse Engineering How Networks Learn Group Operations”. In: *International Conference on Machine Learning (ICML)* (2023). arXiv: [2302.03025](https://arxiv.org/abs/2302.03025).
- [7] Wes Gurnee et al. “Universal Neurons in GPT2 Language Models”. In: (2024). arXiv: [2401.12181](https://arxiv.org/abs/2401.12181).
- [8] Yamini Bansal, Preetum Nakkiran, and Boaz Barak. “Revisiting Model Stitching to Compare Neural Representations”. In: *Advances in Neural Information Processing Systems (NeurIPS)* (2021). arXiv: [2106.07682](https://arxiv.org/abs/2106.07682).
- [9] Simon Kornblith et al. “Similarity of Neural Network Representations Revisited”. In: *International Conference on Machine Learning (ICML)* (2019). arXiv: [1905.00414](https://arxiv.org/abs/1905.00414).
- [10] Minyoung Huh et al. “The Platonic Representation Hypothesis”. In: *International Conference on Machine Learning (ICML)* (2024). arXiv: [2405.07987](https://arxiv.org/abs/2405.07987).
- [11] V. I. Lomonosov. “Invariant subspaces for the family of operators which commute with a completely continuous operator”. In: *Functional Analysis and Its Applications* 7.3 (1973), pp. 213–214. DOI: [10.1007/BF01080698](https://doi.org/10.1007/BF01080698).
- [12] Per Enflo. “On the invariant subspace problem for Banach spaces”. In: *Acta Mathematica* 158 (1987), pp. 213–313. DOI: [10.1007/BF02392260](https://doi.org/10.1007/BF02392260).
- [13] Bernard Beauzamy. *Introduction to Operator Theory and Invariant Subspaces*. Vol. 42. North-Holland Mathematical Library. North-Holland, 1988.

- [14] N. Aronszajn and K. T. Smith. “Invariant subspaces of completely continuous operators”. In: *Annals of Mathematics* 60 (1954), pp. 345–350.
- [15] Marah Abdin et al. “Phi-4 Technical Report”. In: (2024). arXiv: [2412.08905](#).
- [16] Qwen Team. “Qwen3 Technical Report”. In: (2025). arXiv: [2505.09388](#).
- [17] Nelson Elhage et al. “Toy Models of Superposition”. In: *Transformer Circuits Thread* (2022). arXiv: [2209.10652](#).
- [18] David Marr. *Vision: A Computational Investigation into the Human Representation and Processing of Visual Information*. San Francisco: W. H. Freeman, 1982.
- [19] Shreya Saxena and John P. Cunningham. “Towards the neural population doctrine”. In: *Current Opinion in Neurobiology* 55 (2019), pp. 103–111.
- [20] John O’Keefe and Jonathan Dostrovsky. “The hippocampus as a spatial map. Preliminary evidence from unit activity in the freely-moving rat”. In: *Brain Research* 34.1 (1971), pp. 171–175.
- [21] Torkel Hafting et al. “Microstructure of a spatial map in the entorhinal cortex”. In: *Nature* 436 (2005), pp. 801–806.
- [22] Juan A. Gallego et al. “Neural Manifolds for the Control of Movement”. In: *Neuron* 94.5 (2017), pp. 978–984.
- [23] Daniel L. K. Yamins and James J. DiCarlo. “Using goal-driven deep learning models to understand sensory cortex”. In: *Nature Neuroscience* 19.3 (2016), pp. 356–365.
- [24] W. Ross Ashby. *An Introduction to Cybernetics*. London: Chapman & Hall, 1956.
- [25] Karl Friston. “The free-energy principle: a unified brain theory?” In: *Nature Reviews Neuroscience* 11.2 (2010), pp. 127–138.
- [26] Alessandro Achille and Stefano Soatto. “Emergence of Invariance and Disentanglement in Deep Representations”. In: *Journal of Machine Learning Research* 19.50 (2018), pp. 1–34.
- [27] Charles J. Read. “A solution to the invariant subspace problem on the space ℓ_1 ”. In: *Bulletin of the London Mathematical Society* 17.4 (1985), pp. 305–317.
- [28] Spiros A. Argyros and Richard G. Haydon. “A hereditarily indecomposable $\mathcal{L}_\infty$ -space that solves the scalar-plus-compact problem”. In: *Acta Mathematica* 206.1 (2011), pp. 1–54.
- [29] Nelson Elhage et al. “A Mathematical Framework for Transformer Circuits”. In: *Transformer Circuits Thread* (2021).
- [30] Catherine Olsson et al. “In-context Learning and Induction Heads”. In: *Transformer Circuits Thread* (2022). arXiv: [2209.11895](#).
- [31] Kevin Ro Wang et al. “Interpretability in the Wild: a Circuit for Indirect Object Identification in GPT-2 Small”. In: *International Conference on Learning Representations (ICLR)* (2023). arXiv: [2211.00593](#).
- [32] Mor Geva et al. “Transformer Feed-Forward Layers Are Key-Value Memories”. In: *Empirical Methods in Natural Language Processing (EMNLP)* (2021). arXiv: [2012.14913](#).

- [33] Kevin Meng et al. “Locating and Editing Factual Associations in GPT”. In: *Advances in Neural Information Processing Systems (NeurIPS)* (2022). arXiv: [2202.05262](#).
- [34] Arthur Conmy et al. “Towards Automated Circuit Discovery for Mechanistic Interpretability”. In: *Advances in Neural Information Processing Systems (NeurIPS)* (2023). arXiv: [2304.14997](#).
- [35] Tom Lieberum et al. “Gemma Scope: Open Sparse Autoencoders Everywhere All At Once on Gemma 2”. In: (2024). arXiv: [2408.05147](#).
- [36] Samuel Marks et al. “Sparse Feature Circuits: Discovering and Editing Interpretable Causal Graphs in Language Models”. In: (2024). arXiv: [2403.19647](#).
- [37] Neel Nanda et al. “Progress measures for grokking via mechanistic interpretability”. In: *International Conference on Learning Representations (ICLR)* (2023). arXiv: [2301.05217](#).
- [38] Tomas Mikolov et al. “Distributed Representations of Words and Phrases and their Compositionality”. In: *Advances in Neural Information Processing Systems (NeurIPS)* (2013). arXiv: [1310.4546](#).
- [39] Samuel L. Smith et al. “Offline bilingual word vectors, orthogonal transformations and the inverted softmax”. In: *International Conference on Learning Representations (ICLR)* (2017). arXiv: [1702.03859](#).
- [40] Alexis Conneau et al. “Word Translation Without Parallel Data”. In: *International Conference on Learning Representations (ICLR)* (2018). arXiv: [1710.04087](#).
- [41] Maithra Raghu et al. “SVCCA: Singular Vector Canonical Correlation Analysis for Deep Learning Dynamics and Interpretability”. In: *Advances in Neural Information Processing Systems (NeurIPS)* (2017). arXiv: [1706.05806](#).
- [42] Ari S. Morcos, Maithra Raghu, and Samy Bengio. “Insights on representational similarity in neural networks with canonical correlation”. In: *Advances in Neural Information Processing Systems (NeurIPS)* (2018). arXiv: [1806.05759](#).
- [43] Alex H. Williams et al. “Generalized Shape Metrics on Neural Representations”. In: *Advances in Neural Information Processing Systems (NeurIPS)* (2021). arXiv: [2110.14739](#).
- [44] Max Klabunde et al. “Similarity of Neural Network Models: A Survey of Functional and Representational Measures”. In: (2024). arXiv: [2305.06329](#).
- [45] Karel Lenc and Andrea Vedaldi. “Understanding image representations by measuring their equivariance and equivalence”. In: *Conference on Computer Vision and Pattern Recognition (CVPR)* (2015). arXiv: [1411.5908](#).
- [46] Norbert Wiener. *Cybernetics: Or Control and Communication in the Animal and the Machine*. Cambridge, MA: MIT Press, 1948.
- [47] Humberto R. Maturana and Francisco J. Varela. *Autopoiesis and Cognition: The Realization of the Living*. Dordrecht: D. Reidel, 1980.
- [48] Naftali Tishby and Noga Zaslavsky. “Deep Learning and the Information Bottleneck Principle”. In: *IEEE Information Theory Workshop (ITW)* (2015). arXiv: [1503.02406](#).

- [49] Sepp Hochreiter and Jürgen Schmidhuber. “Long Short-Term Memory”. In: *Neural Computation* 9.8 (1997), pp. 1735–1780.
- [50] Albert Gu, Karan Goel, and Christopher Ré. “Efficiently Modeling Long Sequences with Structured State Spaces”. In: *International Conference on Learning Representations (ICLR)* (2022). arXiv: [2111.00396](#).
- [51] Albert Gu and Tri Dao. “Mamba: Linear-Time Sequence Modeling with Selective State Spaces”. In: (2023). arXiv: [2312.00752](#).
- [52] Ashish Vaswani et al. “Attention Is All You Need”. In: *Advances in Neural Information Processing Systems (NeurIPS)* (2017). arXiv: [1706.03762](#).
- [53] Brenden M. Lake and Marco Baroni. “Generalization without systematicity: On the compositional skills of sequence-to-sequence recurrent networks”. In: *International Conference on Machine Learning (ICML)* (2018). arXiv: [1711.00350](#).
- [54] Dieuwke Hupkes et al. “Compositionality decomposed: how do neural networks generalise?” In: *Journal of Artificial Intelligence Research* 67 (2020), pp. 757–795.
- [55] Nouha Dziri et al. “Faith and Fate: Limits of Transformers on Compositionality”. In: *Advances in Neural Information Processing Systems (NeurIPS)* (2023). arXiv: [2305.18654](#).
- [56] Peter H. Schönemann. “A generalized solution of the orthogonal Procrustes problem”. In: *Psychometrika* 31.1 (1966), pp. 1–10. DOI: [10.1007/BF02289451](#).
- [57] Francesco Mezzadri. “How to generate random matrices from the classical compact groups”. In: *Notices of the AMS* 54.5 (2007), pp. 592–604. arXiv: [math-ph/0609050](#).
- [58] John Hewitt and Christopher D. Manning. “A Structural Probe for Finding Syntax in Word Representations”. In: *North American Chapter of the Association for Computational Linguistics (NAACL)* (2019).
- [59] Yonatan Belinkov. “Probing Classifiers: Promises, Shortcomings, and Advances”. In: *Computational Linguistics* 48.1 (2022), pp. 207–219.
- [60] Ross Taylor et al. “Galactica: A Large Language Model for Science”. In: (2022). arXiv: [2211.09085](#).